Experiment – 4:

Verification of System Properties: Linearity, Time Invariance, Causality and Stability

Objective

(i) To verify the linearity and time-invariance properties of continuous-time and discrete-time systems using suitable input signals and system responses.
(ii) To analyze and classify continuous-time and discrete-time systems as causal or non-causal based on their input-output relationships.
(iii) To analyze and classify systems as memoryless or dynamic based on their dependence on present and past input values.
(iv) To verify the BIBO stability of given systems and classify them as stable or unstable for real-time automation applications.

i) To verify the linearity and time-invariance properties of continuous-time and discrete time systems using suitable input signals and system responses.

Expected Outcomes

- Determine whether the selected system satisfies homogeneity and additivity.
- Verify whether delaying the input produces the corresponding output delay.
- Distinguish linear/time-invariant and non-linear/time-varying behavior.
- Support the conclusion using plots and numerical error.

Software / Laboratory Environment

MATLAB

Formulae / Analysis

Linearity: T{a x1 + b x2}=aT{x1}+bT{x2}.
Homogeneity: T{a x}=aT{x}.
Additivity: T{x1+x2}=T{x1}+T{x2}.
Time invariance: if y(t)=T{x(t)}, then T{x(t−t0)}=y(t−t0).

Experimental Procedure

= Select a representative CT or DT system.
= Generate two independent test inputs.
= Compute the individual system outputs.
= Form a combined input and compare its system output with the corresponding linear combination of outputs
= Delay the input by t0 and compare the system response with the correspondingly delayed original output.
= Calculate numerical differences and classify the system.

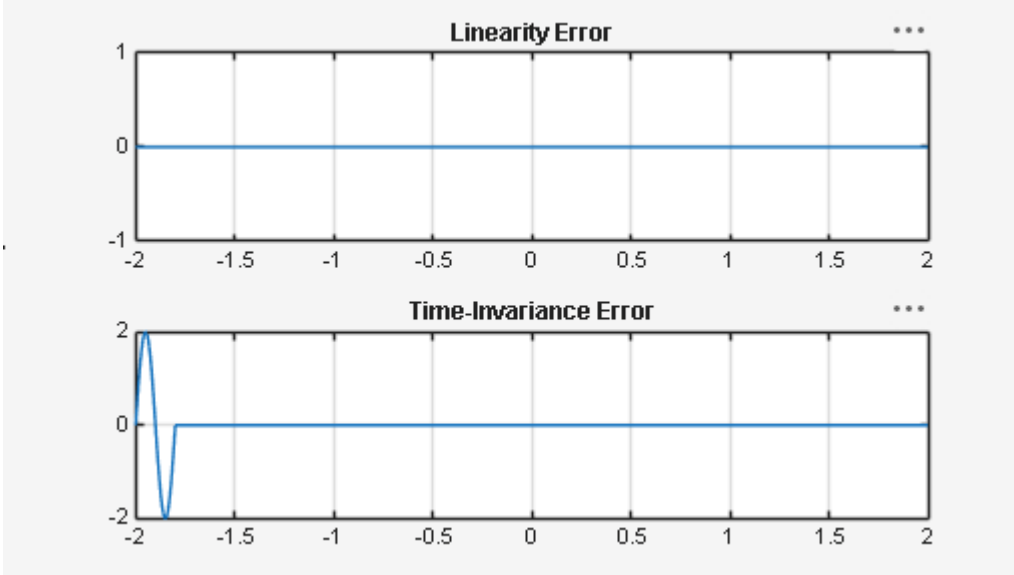
Key Conditions & Threshold Values for Analysis

- Use identical time/sample grids for all comparisons.
- Use the same scaling coefficients in the theoretical and practical tests.
- Allow only small numerical differences due to floating-point computation
- The classification must be based on the stated mathematical conditions.

MATLAB Coding

```
%Experiment 11: Linearity and Time Invariance
clc; clear; close all;
t = -2:0.001:2;
x1 = sin(2*pi*5*t); x2 = cos(2*pi*5*t);
a = 3; b = 5; t0 = 0.2;
T = @(x) 2*x;
lhs = T(a*x1+b*x2);
rhs = a*T(x1)+b*T(x2);
linError = max(abs(lhs-rhs));
xDelay = sin(2*pi*5*(t-t0));
lhsTI = T(xDelay);
rhsTI = T(x1);
```

```
rhsTIshift = interp1(t,rhsTI,t-t0,'linear',0);
tiError = max(abs(lhsTI-rhsTIshift));
figure;
```



Result

The system properties were successfully verified and classified. Linearity, time invariance, causality, memory, and stability were analyzed. The results confirmed the system behavior for practical applications.

4.

Objective	Assessment Criterion	Excellent – 4	Good – 3	Satisfactory – 2	Needs Improvement – 1	Weightage	Marks(10)	Marks obtained
(i) To verify the linearity and time-invariance properties of continuous-time and discrete-time systems using suitable input signals and system responses.	**Linearity Verification (Homogeneity &Additivity)**	Correctly applies the homogeneity test and the additivity test (or the combined superposition test) to every CT and DT system. Chooses suitable input signals and scaling constants, compares the two responses accurately, and concludes linear or non-linear correctly.	Applies the linearity tests correctly to most systems, with minor errors in the choice of inputs, algebra or comparison.	Performs a basic linearity test (only one of homogeneity or additivity, or only some systems) with noticeable errors, and needs guidance.	Unable to apply or interpret the linearity test.	**20**	2	

	Time-Invariance Verification (Time-Shift Test)	Correctly applies the time-shift test, comparing the response to the shifted input with the shifted output for every system. Handles systems with time-scaling or time-varying coefficients correctly and concludes time-invariant or time-varying correctly.	Applies the time-shift test correctly to most systems, with minor errors in shifting or comparison.	Performs a basic time-shift test with noticeable errors (for example shifting only the input or only the output), and needs guidance.	Unable to apply or interpret the time-invariance test.	**20**	2	
	MATLAB Implementation & Numerical Verification	Writes correct MATLAB code with suitable time and index vectors to generate the test inputs, the system responses, the scaled, summed and shifted versions, and the plots.	Code, plots and numerical comparison are correct, with minor errors in range, parameters, tolerance or labelling.	Code runs but has noticeable errors in vectors, shifting or error calculation, and needs assistance.	Unable to implement the tests or produce meaningful responses and plots in MATLAB.	**20**	2	

		Computes numerical errorsbetween the compared responses, and uses a tolerance to decide pass or fail. Plots have titles, axis labels and legends.						
	System Classification & Mathematical Justification	Independently classifies each CT and DT system as linear or non-linear and time-invariant or time-varying Supports each classification with correct mathematical working and with the waveform or error evidence from MATLAB.	Classifies most systems correctly, with minor errors or limited justification.	Gives basic classifications. Justification is incomplete or not linked to the test results, and needs guidance.	Unable to classify the systems correctly or justify the classification.	**20**	2	

	Engineering Analysis, Record & Interpretation	Provides a technically accurate interpretation of the results and links them to the importance of linear time-invariant (LTI) systems Maintains a complete, well-organized record with code, plots, observations and conclusions.	Maintains a proper record with minor omissions. Interpretation is correct but brief.	Record and interpretation are basic or partly incomplete, and need guidance.	Incomplete record, limited or no interpretation of the results.	**20**	2	
					Total	**100**	**0**	